# Reframing preprocessing selection as model-internal calibration in near-infrared spectroscopy

A large-scale benchmark of operator-adaptive PLS and Ridge models

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**Abstract**

Preprocessing screening is often the most expensive part of a near-infrared spectroscopy calibration workflow. It works because smoothing, derivatives, detrending and related filters change the spectral directions seen by PLS or Ridge regression, but a full external search repeatedly refits nearly the same linear model. This paper studies the case where that search can be collapsed into one calibration step.

For strict linear preprocessing operators, the transformed PLS cross-covariance satisfies $(\mathbf{X}\mathbf{A}^\top)^\top\mathbf{Y} = \mathbf{A}\mathbf{X}^\top\mathbf{Y}$, and Ridge regression depends on the operator-induced kernel $\mathbf{X}\mathbf{A}^\top\mathbf{A}\mathbf{X}^\top$. These identities allow a finite operator bank to be screened inside the model while retaining original-wavelength coefficients. Sample-adaptive or fitted corrections such as SNV, MSC, EMSC and ASLS remain fold-local branches, not absorbed into the algebra.

The study uses the AOM benchmark cohort: 61 regression rows and 17 classification rows in the manifest. On the main regression denominator ($N = 32$), plain compact-bank AOM-PLS records median RMSEP ratios of 0.991 against PLS-default and 0.990 against PLS-HPO; the selected `ASLS-AOM-compact-cv5` branch records 0.985 and 1.002 on the same two references. The plain `AOMRidge-global-compact-none` baseline records 0.974 against Ridge-default and 0.984 against Ridge-HPO, while the selected `AOMRidge-Blender-headline-spxy3` records 0.918 and 0.966. The selected classifier, `AOM-PLS-DA-global-simpls-covariance`, improves balanced accuracy by 0.159 on $N = 13$ datasets with 12/13 wins. The runtime gap is the practical result: PLS-HPO takes a median total time of 710.81 s per run, whereas the selected AOM-PLS branch takes 1.63 s. Linear operator-adaptive calibration therefore gives comparable prediction quality to exhaustive preprocessing screening, with orders-of-magnitude less fitting time for PLS.

**Keywords:** near-infrared spectroscopy; chemometrics; preprocessing selection; partial least squares; ridge regression; operator kernels; calibration.

## 1 Introduction

Near-infrared spectroscopy (NIRS) is used because it is rapid, non-destructive and compatible with routine analytical deployment [1, 2]. The calibration step is still fragile. A model that works on raw spectra may fail after a different split, instrument or sample family, while the same regressor with a derivative, smoother or baseline correction can become usable. In practice, much of the model-building effort is therefore spent screening preprocessing choices.

The usual response is external search. A calibration workflow enumerates normalizations, smoothers, derivative orders, baseline corrections and model hyperparameters, then chooses the best validated pipeline. That workflow is straightforward and often effective, but it is slow. It also hides a scientific question inside a large engineering loop: which spectral linear geometry does the calibration model need?

The cost is concrete in this benchmark. The full PLS preprocessing grid evaluates 600 combinations across five trials, and the Ridge grid evaluates the same combinations across ten trials, giving 3000 and 6000 candidate fits per dataset/seed before the final refit. Repeating this search on small calibration sets can also make the selected pipeline unstable: the chosen preprocessing is the winner of a multiplicative loop over folds, operators and model hyperparameters. A single operator-adaptive linear fit is easier to audit because the selected geometry is part of one calibration object with coefficients recovered on the original wavelength axis.

This paper focuses on the part of preprocessing that has a simple algebraic answer. If a preprocessing step is a fixed linear operator on the wavelength axis, it can be represented by a matrix $\mathbf{A}$ and moved inside PLS or Ridge calibration. The model can then screen a bank of such operators in one fit rather than refitting every transformed data matrix as an external pipeline. This does not claim that every preprocessing step is linear. SNV, MSC, EMSC and ASLS are fitted or sample-adaptive transformations; they remain fold-local branches in the experimental protocol.

The contribution is deliberately narrow. We derive the PLS covariance identity and the Ridge operator-kernel identity, instantiate them in AOM-PLS and AOM-Ridge, and test the resulting models against default PLS/Ridge and full cartesian preprocessing-HPO baselines. The question is not whether a new nonlinear learner can win a benchmark. The question is whether the preprocessing search already used with linear chemometric models can be made faster, simpler and easier to audit.

The key empirical check is intentionally plain. The AOM-compact-cv5 PLS row and the AOMRidge-global-compact-none Ridge row use compact strict-linear banks without the selected ASLS branch, Blender selector or local-neighbourhood machinery. If these simple AOM baselines already compete with default and preprocessing-HPO PLS/Ridge, the algebra is doing useful work rather than only moving complexity into another tuning grid.

## 2 Related work

Spectral preprocessing is a central part of chemometrics. Common tools include Savitzky–Golay smoothing and derivatives [3], Norris–Williams derivatives [4], SNV [5], MSC [6], EMSC [7] and penalized baseline smoothers [8, 9]. Reviews emphasize that these transformations remove different nuisance structures and should be chosen with the analytical setting in mind [10].

PLS remains the reference linear method for many NIRS calibration problems [11, 12]. NIPALS and SIMPLS offer two useful views: one iterative and score-based, the other covariance-based [13]. Ridge regression provides a complementary regularized linear model [14]; in dual form it depends on the sample Gram matrix, which makes preprocessing a change of kernel geometry [15, 16].

This work combines these two lines. It keeps the chemometric interpretability of linear calibration, but treats strict preprocessing alternatives as operators that can be selected inside the model rather than as independent outer-loop pipelines.

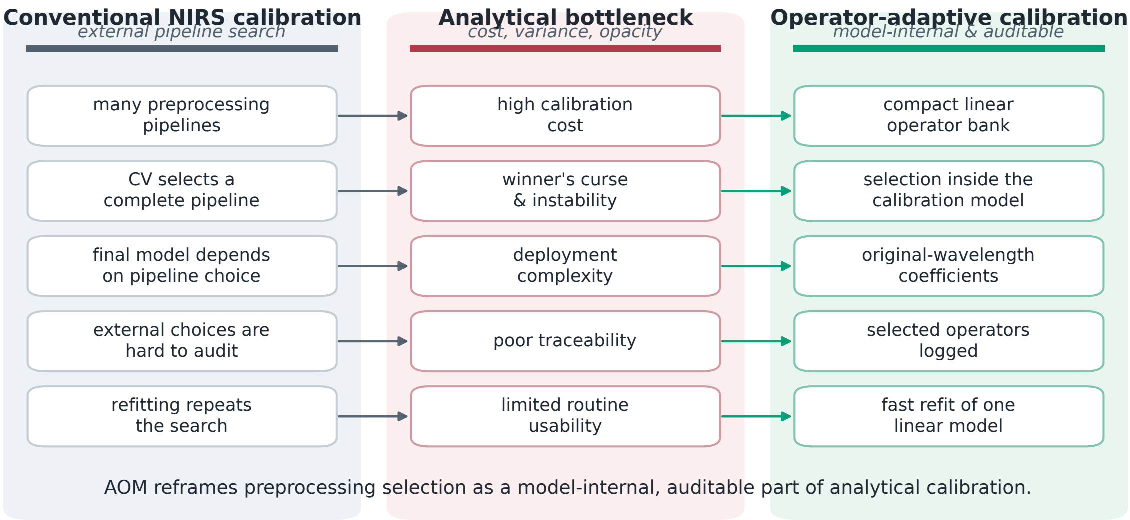


Figure 1: External preprocessing screening repeatedly fits transformed pipelines. Operator-adaptive calibration moves strict linear alternatives inside the calibration model; fitted or sample-adaptive corrections remain fold-local branches.

# 3 Linear-operator scope

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a centered spectral matrix, with rows as samples and columns as wavelengths. A strict linear preprocessing operator is a fixed matrix $\mathbf{A} \in \mathbb{R}^{p \times p}$ applied to each row spectrum:

$$\mathbf{X}_{\mathbf{A}} = \mathbf{X}\mathbf{A}^{\mathsf{T}}. \tag{1}$$

The operator is fixed once the wavelength grid and operator parameters are chosen. It does not depend on the response, the validation fold, the current sample, or a reference spectrum estimated from the calibration set.

The strict-linear bank used in the main experiments contains identity, Savitzky–Golay smoothers, Savitzky–Golay first and second derivatives, finite differences, and polynomial detrending projections. The compact bank is intentionally small: it has nine operators, so internal CV selection evaluates a small number of candidate geometries rather than a large cartesian preprocessing grid.

The outside-scope examples are important. SNV uses sample-specific centering and scaling. MSC and EMSC estimate correction parameters relative to a reference. ASLS estimates a baseline through an iterative asymmetric criterion. These transformations are allowed in the benchmark, but only as branches fitted inside each calibration fold. The selected PLS result uses such a branch followed by the compact strict-linear AOM bank.

# 4 Methods

## 4.1 Covariance identity for one-shot AOM-PLS

For centered $\mathbf{X}$ and $\mathbf{Y}$, the transformed PLS cross-covariance for operator $\mathbf{A}_b$ is

$$(\mathbf{X}\mathbf{A}_b^{\mathsf{T}})^{\mathsf{T}}\mathbf{Y} = \mathbf{A}_b\mathbf{X}^{\mathsf{T}}\mathbf{Y}. \tag{2}$$

Equation 2 is the core simplification. It means that a bank of strict linear preprocessings can be screened by acting on $\mathbf{X}^{\mathsf{T}}\mathbf{Y}$, without fitting one full PLS model for every transformed matrix.

Table 1: Operator-bank scope. Strict linear operators are handled by the AOM identities; fitted or sample-adaptive transformations remain explicit fold-local branches.

| Family | Operators in compact bank | Strict linear status |
|---|---|---|
| Identity | identity transform | yes |
| Smoothing | Savitzky–Golay smoothing, windows 11 and 21 | yes |
| Derivatives | Savitzky–Golay first derivative, windows 11 and 21 | yes |
| Derivatives | Savitzky–Golay second derivative, window 11 | yes |
| Baseline trend | polynomial detrending, degrees 1 and 2 | yes, fitted inside calibration folds |
| Local contrast | first finite difference | yes |
| Scatter correction | SNV, MSC, EMSC | branch preprocessing, fold-local |
| Baseline correction | ASLS and related asymmetric smoothers | branch preprocessing, fold-local |

The simulation check for this identity is direct: explicit transformed-space PLS and covariance-space AOM-PLS must select the same operator direction when the same centered fold and operator bank are supplied. The check is used as an implementation guard, while the benchmark below evaluates the effect on held-out prediction.

### 4.2 AOM-PLS solvers and coefficient recovery

A standard PLS coefficient matrix can be written as

$$\mathbf{B} = \mathbf{W}(\mathbf{P}^{\mathsf{T}}\mathbf{W})^{+}\mathbf{Q}^{\mathsf{T}}, \tag{3}$$

where $\mathbf{W}$ are weights, $\mathbf{P}$ are spectral loadings, $\mathbf{Q}$ are response loadings and $+$ denotes the Moore–Penrose inverse. In AOM-PLS, a component may be selected in a transformed space. If the transformed-space direction is $r_a$ under operator $\mathbf{A}_b$, its original-wavelength direction is

$$z_a = \mathbf{A}_b^{\mathsf{T}} r_a. \tag{4}$$

With $\mathbf{Z} = [z_1, \ldots, z_k]$ and scores $\mathbf{T} = \mathbf{XZ}$, the recovered coefficient matrix is

$$\mathbf{B} = \mathbf{Z}(\mathbf{P}^{\mathsf{T}}\mathbf{Z})^{+}\mathbf{Q}^{\mathsf{T}}. \tag{5}$$

The fitted model is therefore still one linear calibration on the original wavelength grid.

Two solver views are implemented. The NIPALS-adjoint solver applies operator adjoints during iterative score extraction. The SIMPLS-covariance solver uses Equation 2 directly in covariance space. The main plain baseline is `AOM-compact-cv5`: the compact nine-operator bank with internal five-fold selection and no fitted ASLS branch. The selected regression result uses the same compact CV5 branch after ASLS is fitted fold-locally. The default bank, SIMPLS-only variants, and other non-selected AOM variants are reported in the supplementary material.

The compact bank was chosen from the selector diagnostics, not by adding an unbounded operator catalogue. In the refreshed diagnostics, the three most frequent strict operators (`sg_smooth_w21_p3`, `sg_d1_w21_p3` and `sg_d1_w11_p2`) account for 54.7% of compact-bank component selections, while identity remains selected on 8.7% of components. The operator bank is therefore diverse enough to adapt, but small enough to keep the selection problem stable.

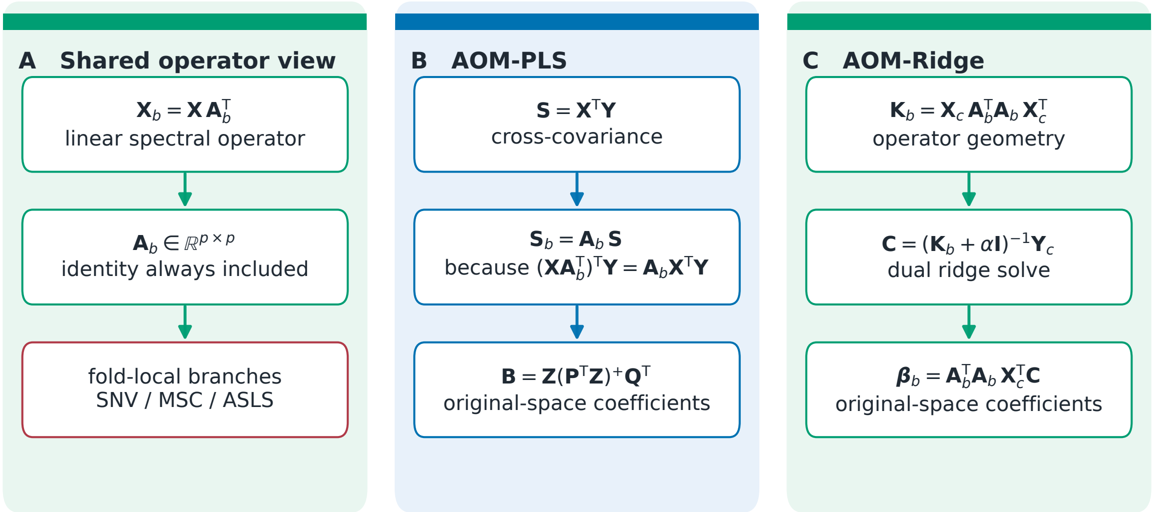


Figure 2: AOM mathematical structure. AOM-PLS screens operators through cross-covariances; AOM-Ridge screens operator-induced kernels. Both recover coefficients on the original wavelength grid.

### 4.3 AOM-Ridge operator kernels

For centered spectra $\mathbf{X}_c$ and response $\mathbf{Y}_c$, Ridge regression can be solved in the dual. Operator $\mathbf{A}_b$ induces the kernel

$$\mathbf{K}_b = \mathbf{X}_c\mathbf{A}_b^{\mathsf{T}}\mathbf{A}_b\mathbf{X}_c^{\mathsf{T}}. \tag{6}$$

For regularization $\alpha > 0$,

$$\mathbf{C} = (\mathbf{K}_b + \alpha\mathbf{I}_n)^{-1}\mathbf{Y}_c, \qquad \boldsymbol{\beta}_b = \mathbf{A}_b^{\mathsf{T}}\mathbf{A}_b\mathbf{X}_c^{\mathsf{T}}\mathbf{C}. \tag{7}$$

Weighted operator mixtures are obtained by replacing $\mathbf{K}_b$ with

$$\mathbf{K} = \sum_b s_b^2\mathbf{X}_c\mathbf{A}_b^{\mathsf{T}}\mathbf{A}_b\mathbf{X}_c^{\mathsf{T}}, \tag{8}$$

which gives an original-space coefficient matrix through the matching operator metric $\sum_b s_b^2\mathbf{A}_b^{\mathsf{T}}\mathbf{A}_b$.

The plain Ridge baseline is `AOMRidge-global-compact-none`: one global selection over the compact strict-linear bank, without Blender mixing or local neighbourhood routing. The selected Ridge result uses the Blender selector (`AOMRidge-Blender-headline-spxy3`). It combines several strong operator geometries through internal validation rather than committing to one noisy winner. GCV, local-neighbourhood and additional global compact selectors are mathematically described in the supplement.

## 5 Datasets and protocol

### 5.1 Simulation checks

The implementation checks use explicit-vs-operator simulations for the PLS covariance identity and the Ridge kernel identity. These checks verify that for fixed folds and fixed strict operators, the folded computation matches the explicit transformed computation used by an external search.

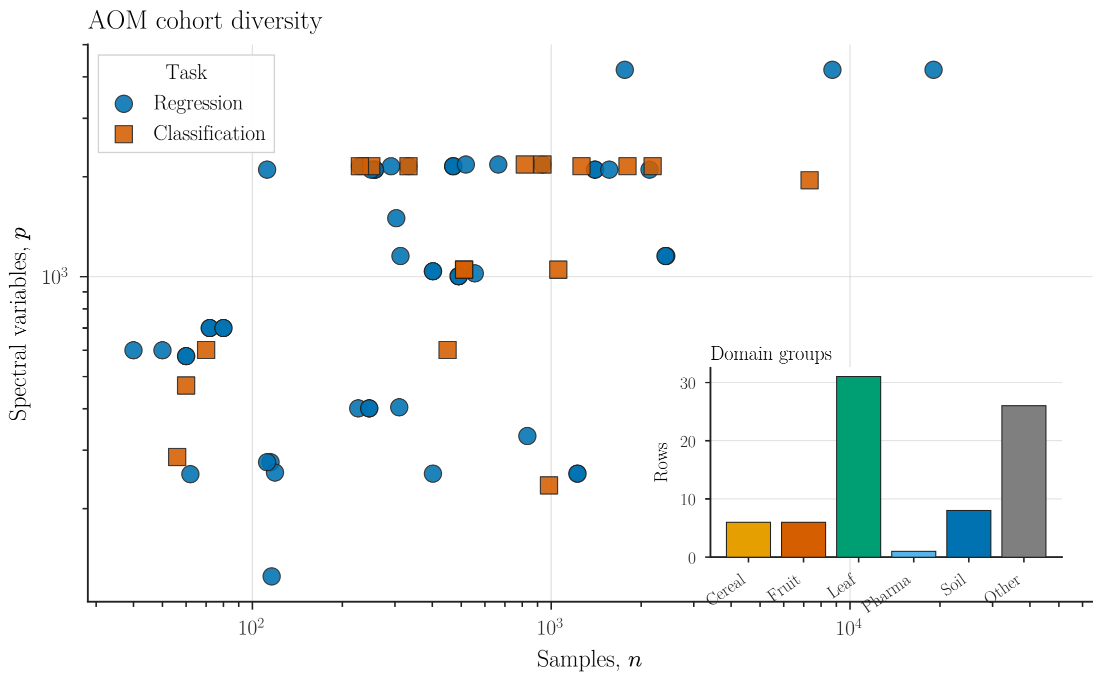


Figure 3: Dataset diversity in the AOM cohort. Points show total sample count and number of spectral variables on logarithmic axes; colours separate regression and classification rows, and the inset summarizes coarse application domains.

Table 2: Dataset shape summary for the AOM cohort. Here $n$ is the total calibration plus external-test sample count, $p$ is the number of spectral variables, $C$ is the number of classes and $I$ is the largest-class share for classification rows.

| Task | $N$ | $n_{\mathrm{median}}$ | $n_{\mathrm{min}}$ | $n_{\mathrm{max}}$ | $p_{\mathrm{median}}$ | $p_{\mathrm{min}}$ | $p_{\mathrm{max}}$ | $(p/n)_{\mathrm{median}}$ | $C_{\mathrm{median}}$ | $I_{\mathrm{median}}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| Classification | 17 | 511 | 56 | 7323 | 1951 | 235 | 2177 | 2.055 | 2 | 0.513 |
| Regression | 61 | 402 | 40 | 45417 | 1023 | 125 | 4200 | 2.382 | – | – |

## 5.2 Datasets

The benchmark cohort is intentionally heterogeneous. It combines public and local NIRS calibration tasks from leaf physiology, fruit quality, grain and seed traits, dairy, beverages, meat quality, petroleum, soil amendments, wood products and pharmaceutical tablets. Several source databases contribute more than one response variable, so a row denotes a concrete prediction or classification task rather than only a spectral database. The cohort covers small calibration settings, high-sample leaf datasets and several high-dimensional low-$n$ regimes.

Table 2 summarizes the cohort in the same compact form used for the companion benchmark inventory. Regression contains 61 rows with median $n = 402$ samples and median $p = 1023$ spectral variables. The classification manifest contains 17 rows, of which 16 are included in the benchmark analyses after one missing-file exclusion. Across the 17 manifest rows, the median $n$ is 511 and the median $p$ is 1951; the median number of classes is two and the median largest-class share is 0.513. The complete per-dataset manifest description is provided in the supplementary longtable, while all score tables below use their explicitly stated paired denominators.

Whenever a source dataset already defined an external split, that split was preserved. Otherwise deterministic SPXY was used for regression, with the stratified variant used for classification when class labels were present [17, 18]. Test sets were kept outside preprocessing, operator, component and regularization selection. No explicit outlier removal was applied during

Table 3: Cohort denominators and data-quality rules used in the main manuscript.

| Property | Summary used in this work |
|---|---|
| Regression manifest | 61 included regression rows across the benchmark inventory. |
| Classification manifest | 17 classification manifest rows; 16 included rows after one missing-file exclusion. |
| Main regression denominator | $N_\cap = 32$ datasets for the eight paper variants. |
| Paper variants | PLS-default, PLS-HPO, AOM-PLS (simple), AOM-PLS (best), Ridge-default, Ridge-HPO, AOM-Ridge (simple) and AOM-Ridge (best). |
| Analytical domains | Leaf physiology, fruit quality, grain and seed traits, dairy, beverages, meat quality, petroleum, soil, manure, wood products, tablets and public calibration datasets. |
| Sample and wavelength range | Training sets span 28–39,225 samples and 125–4,200 spectral variables in the local regression manifest. |
| Validation rule | External test sets are never used for preprocessing, operator, component or regularization selection. |

cohort construction: calibration and test samples were kept as supplied by the source datasets.

### 5.3 Benchmark protocol

The empirical benchmark reports eight regression variants and one selected classification variant. All paired comparisons reported below are computed on the strict intersection $N_\cap = 32$ of cohort rows available for every paper variant, so that ratios and win counts are directly comparable across variant pairs. The classification cohort contains 17 manifest rows, with 13 paired datasets for the selected classifier.

All selection steps are internal to the calibration set. RMSEP is the primary regression metric. Because response scales differ across traits, aggregate regression results use paired RMSEP ratios. Classification uses balanced accuracy. Paired tests use one-sided Wilcoxon signed-rank tests with Holm correction within the displayed family.

The cartesian-HPO baselines are intentionally strong and expensive. They enumerate 600 preprocessing combinations. PLS-HPO uses five trials per combination; Ridge-HPO uses ten trials per combination.

The eight paper labels correspond to the following runner keys: PLS-default (`pls-default-cv5`), PLS-HPO (`pls-tabpfn-hpo-25trials`), AOM-PLS (simple) (`AOM-compact-cv5`), AOM-PLS (best) (`ASLS-AOM-compact-cv5`), Ridge-default (`ridge-default-cv5`), Ridge-HPO (`ridge-tabpfn-hpo-60trials`), AOM-Ridge (simple) (`AOMRidge-global-compact-none`) and AOM-Ridge (best) (`AOMRidge-Blender-headline-spxy3`).

## 6 Results

### 6.1 PLS regression

The plain AOM-PLS regression baseline is `AOM-compact-cv5`. Against PLS-default, it reaches a median RMSEP ratio of 0.991 on $N = 32$ datasets with 22/32 wins. Against PLS-HPO, it is near parity, with a median ratio of 0.990 on $N = 32$ and 19/32 wins. This row has no ASLS branch and no enlarged operator search; it is the compact strict-linear bank selected by internal five-fold validation.

The selected AOM-PLS regression variant is `ASLS-AOM-compact-cv5`. It is compared in the main text with the pre-registered linear references. Against PLS-default, it reaches a median

Table 4: Main paired regression results. Ratios below one favour the row method over the reference named in the comparison.

| Comparison | Evidence source | $N$ | Median RMSEP ratio / wins; $p_{\mathrm{Holm}}$ |
|---|---|---|---|
| AOM-PLS (simple) vs PLS-default | AOM-PLS seeds012 / default-CV all | 32 | 0.991; 22/32; 0.896 |
| AOM-PLS (best) vs PLS-default | AOM-PLS seeds012 / default-CV all | 32 | 0.985; 20/32; 1.000 |
| AOM-PLS (simple) vs PLS-HPO | AOM-PLS seeds012 / cartesian HPO seeds012 | 32 | 0.990; 19/32; 1.000 |
| AOM-PLS (best) vs PLS-HPO | AOM-PLS seeds012 / cartesian HPO seeds012 | 32 | 1.002; 15/32; 1.000 |
| PLS-HPO vs PLS-default | cartesian HPO seeds012 / default-CV all | 32 | 0.992; 19/32; 1.000 |
| Ridge-HPO vs Ridge-default | cartesian HPO seeds012 / default-CV all | 32 | 0.962; 19/32; 1.000 |
| AOM-Ridge (simple) vs Ridge-default | AOM-Ridge headline / default-CV all | 32 | 0.974; 25/32; 0.007 |
| AOM-Ridge (best) vs Ridge-default | AOM-Ridge headline / default-CV all | 32 | 0.918; 27/32; 2.6e-04 |
| AOM-Ridge (simple) vs Ridge-HPO | AOM-Ridge headline / cartesian HPO seeds012 | 32 | 0.984; 19/32; 1.000 |
| AOM-Ridge (best) vs Ridge-HPO | AOM-Ridge headline / cartesian HPO seeds012 | 32 | 0.966; 25/32; 0.033 |

RMSEP ratio of 0.985 on $N = 32$ datasets with 20/32 wins. Against PLS-HPO, it is essentially tied: median ratio 1.002 on $N = 32$, 15/32 wins.

This is the intended interpretation. Full external preprocessing-HPO can recover the benefit of careful preprocessing, but the gain is obtained by repeatedly fitting the linear calibration under a much larger search budget. AOM-PLS makes the strict-linear part of the same decision inside one coefficient-bearing calibration object.

### 6.2 Ridge regression

The plain Ridge AOM baseline is `AOMRidge-global-compact-none`. It improves over default Ridge with a median RMSEP ratio of 0.974 on $N = 32$ datasets, 25/32 wins and Holm-adjusted $p = 0.007$. Against Ridge-HPO, it remains close to the grid baseline: median ratio 0.984 on $N = 32$, 19/32 wins.

The selected Ridge variant is `AOMRidge-Blender-headline-spxy3`. It improves strongly over default Ridge: median RMSEP ratio 0.918 on $N = 32$ datasets with 27/32 wins and Holm-adjusted $p = 2.6 \times 10^{-4}$. It also improves over Ridge-HPO, with median ratio 0.966 on $N = 32$, 25/32 wins and $p = 0.033$.

The Ridge result supports the operator-kernel view. In Ridge, preprocessing changes the regularized sample geometry. Blender is the best main selector because it combines strong operator geometries through internal validation while preserving a single deployable Ridge model.

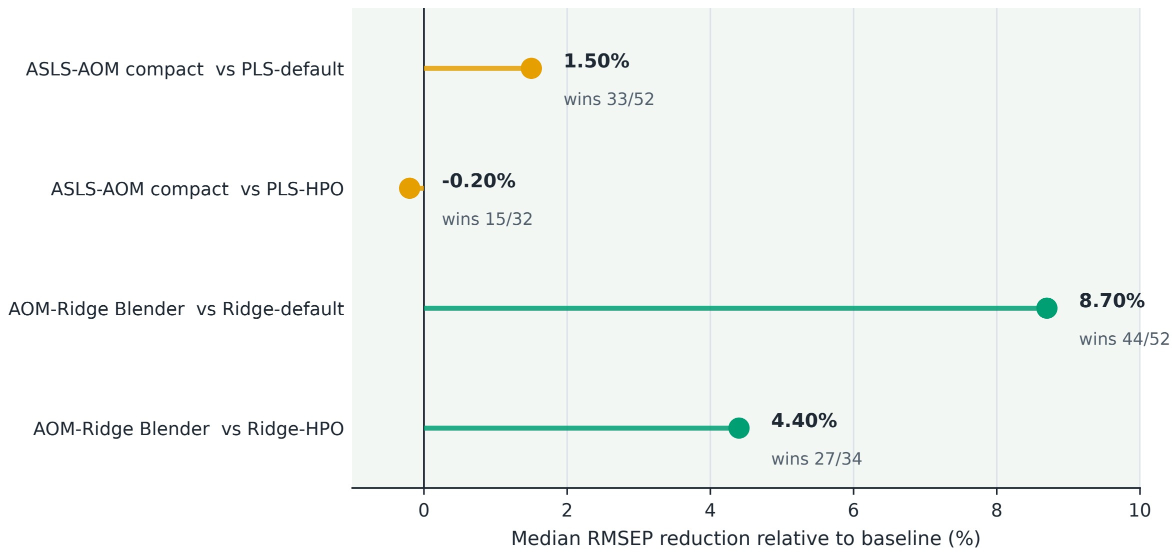


Figure 4: Eight paired regression comparisons for the plain compact-bank AOM baselines and the selected AOM-PLS and AOM-Ridge variants against their PLS-default, PLS-HPO, Ridge-default and Ridge-HPO references.

Table 5: Main classification result. Positive balanced-accuracy differences favour the operator-adaptive classifier.

| Comparison | $N$ | Median $\Delta$ balanced acc. | 95% CI | Wins; $p_{\mathrm{Holm}}$ |
|---|---|---|---|---|
| AOM-PLS-DA-global-simpls-covariance vs PLS-DA | 13 | 0.159 | 0.129–0.422 | 12/13; 0.007 |

## 6.3 Classification

The selected classification variant is `AOM-PLS-DA-global-simpls-covariance`. On the paired classification cohort, it improves balanced accuracy by a median 0.159 on $N = 13$ datasets, with 12/13 wins and Holm-adjusted $p = 0.007$. This secondary result confirms that the covariance-space operator selection used for regression transfers to PLS-DA when the response encoding and scoring metric are changed.

## 6.4 Time budget

The practical advantage is runtime. PLS-HPO has a median total time of 710.81 s per run and an observed row total of 40.8 h. AOM-PLS (simple), implemented as `AOM-compact-cv5`, has a median total time of 1.18 s, and AOM-PLS (best) has a median total time of 1.63 s and an observed total of 0.1 h. On median total time, AOM-PLS (best) is about 436 times faster than PLS-HPO. In runner-level search units, the contrast is 3000 trial fits per PLS-HPO dataset/seed versus 45 compact-bank CV fits for each AOM-PLS row.

Ridge is different because AOM-Ridge (best), the Blender selector, is heavier. It still improves over Ridge-HPO, and its median total time is 728.81 s compared with 1584.00 s for Ridge-HPO. AOM-Ridge (simple), implemented as `AOMRidge-global-compact-none`, is much lighter, with median total time 23.78 s and 450 operator-alpha cells. The result is not that every AOM selector is instant; it is that strict-linear preprocessing can be represented as a model geometry rather than as a blind external pipeline loop.

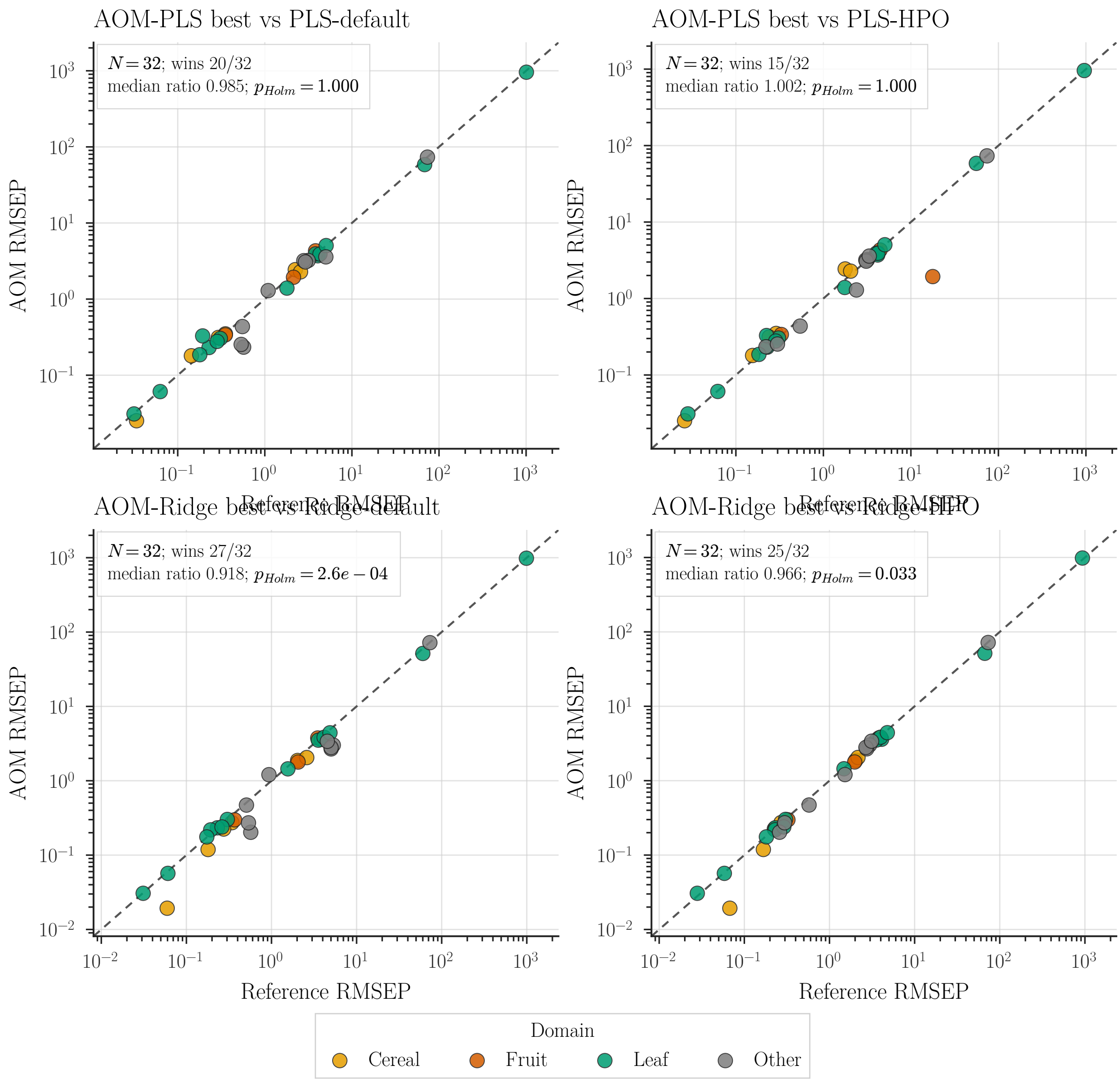


Figure 5: Prediction-quality scatter plots for the selected AOM-PLS and AOM-Ridge variants against default and HPO references. Axes are log RMSEP; points below the diagonal favour AOM.

# 7 Discussion

The benchmark supports a simple engineering conclusion: when preprocessing alternatives are strict linear operators, screening them outside the model is often unnecessary. AOM-PLS folds the operator choice into the PLS cross-covariance calculation, and AOM-Ridge folds it into the Ridge kernel. The resulting calibration remains linear and auditable.

The strongest use case is the one common in NIRS: small to moderate calibration sets, noisy validation folds and many plausible preprocessing choices. In that regime, an exhaustive external search spends most of its time refitting variations of the same linear model. The grid baselines here require 3000 PLS or 6000 Ridge candidate fits per dataset/seed, whereas the compact AOM-PLS selector evaluates 45 operator/component candidates. The observed median total times reflect that difference: 710.81 s for PLS-HPO against 1.18–1.63 s for the compact AOM-PLS rows.

The statistical reason is related to the engineering cost. A large cartesian grid repeatedly

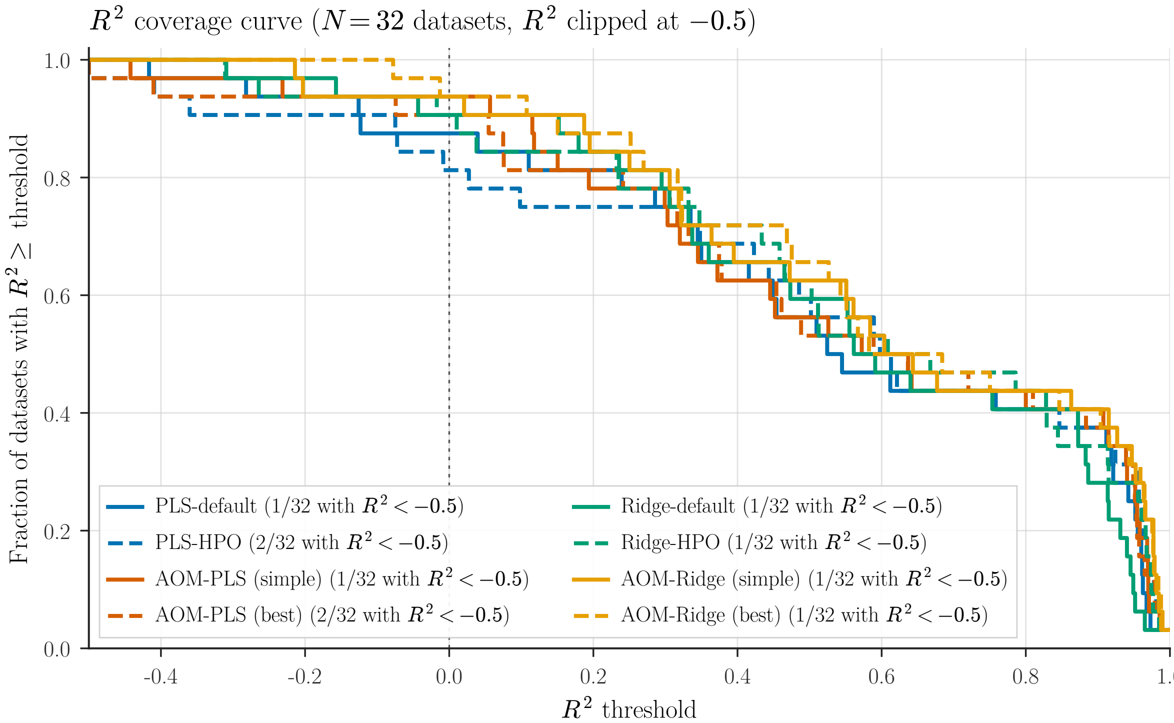


Figure 6: $R^2$ coverage curves for the eight paper variants on the main regression denominator. Higher curves indicate a larger fraction of datasets exceeding each test-$R^2$ threshold.

re-estimates pipelines on small folds and then promotes the winner of that noisy search. The strict-linear AOM step instead selects operators from a covariance or kernel representation of one calibration problem. It does not remove validation uncertainty, but it avoids turning the choice of a derivative or smoother into a multiplicative refitting loop.

Auditability and re-deployment are also different. A grid winner is an external preprocessing pipeline plus a downstream linear model; every fitted branch must be reproduced correctly when the calibration is moved to another instrument or software stack. AOM-PLS and AOM-Ridge recover coefficients on the original wavelength axis (Equations 5 and 7), so the deployed object is still one linear calibration whose spectral coefficients can be inspected directly.

The plain baselines are the most useful evidence for choosing this route over a preprocessing grid. `AOM-compact-cv5` already improves modestly over default PLS and reaches near-parity with PLS-HPO, while `AOMRidge-global-compact-none` improves over default Ridge and remains close to Ridge-HPO. The selected ASLS-AOM and Blender rows add accuracy, but the compact no-frills rows show that the main gain is not dependent on a large new tuning apparatus.

The scope is also clear. AOM algebra does not absorb sample-adaptive transforms. SNV, MSC, EMSC and ASLS must be fitted inside folds. This is why the selected PLS result is described as an ASLS branch followed by strict linear AOM, not as ASLS being part of the operator identity. When such branches dominate a dataset, they should remain explicit and audited.

Taken together, the simple-baseline result is the practical summary: before adding the selected ASLS or Blender refinements, compact AOM already captures much of the value of preprocessing search while keeping the model linear, fast and directly deployable.

Table 6: Search and runtime budget for the main methods. Observed totals are sums over benchmark rows.

| Method | Datasets | Search budget | Median fit (s) | Median total (s) |
|---|---|---|---|---|
| PLS-default | 32 | 25 component trials; observed 0.1 h | 0.02 | 1.21 |
| PLS-HPO | 32 | 600 x 5 trials; observed 40.8 h | 0.03 | 710.81 |
| AOM-PLS (simple) | 32 | 9 operators x CV-5; observed 0.1 h | 1.18 | 1.18 |
| AOM-PLS (best) | 32 | ASLS branch + 9 operators x CV-5; observed 0.1 h | 1.43 | 1.63 |
| Ridge-default | 32 | 15 alpha trials; observed 0.0 h | 0.00 | 0.38 |
| Ridge-HPO | 32 | 600 x 10 trials; observed 96.5 h | 0.05 | 1584.00 |
| AOM-Ridge (simple) | 32 | 9 operators x 50 alpha cells; observed 0.5 h | 23.77 | 23.78 |
| AOM-Ridge (best) | 32 | AOM-Ridge Blender: 8 candidates x 3 outer folds + 8 refits; observed 9.3 h | 727.51 | 728.81 |

# 8 Conclusion

Preprocessing screening for linear NIRS calibration works, but it is often implemented as an expensive outer loop. For strict linear operators, PLS and Ridge provide identities that move the operator choice inside the calibration itself. On the available AOM benchmark, the selected AOM-PLS and AOM-Ridge variants are comparable to or better than full preprocessing-HPO on the reported paired tests, with a large time advantage for PLS. The useful scientific message is therefore modest and practical: keep fitted corrections fold-local, fold strict linear operators into the model, and report every comparison on its paired denominator.

# Data and code availability

The public software entry point for the methods, benchmark runners, result tables and manuscript artifacts reported here is `nirs4all-aom` [19] at `https://github.com/GBeurier/nirs4all-aom`. The `nirs4all` project [20] is cited only for the NIRS instrumentation, acquisition and provenance context used to assemble local benchmark inputs; it is not the software entry point for the AOM methods reported in this paper.

The reproducibility protocol is defined by the cohort denominators in Section 5, the fold-local model-selection rules in the Methods, and paired dataset-level tests with Holm correction for the reported families of comparisons. The supplementary material gives the full cohort manifest, per-dataset regression table, prediction-quality diagnostics, seed-stability diagnostics and software-validation matrix. Table 7 summarizes the software artifacts and validation status for the public release.

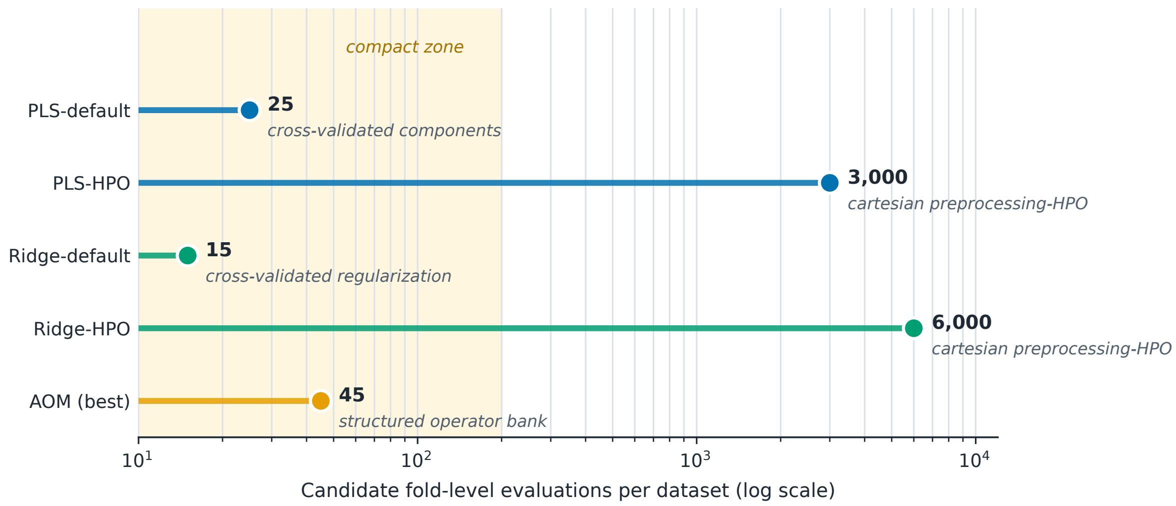


Figure 7: Search-budget scale for the eight paper variants: PLS-default, PLS-HPO, AOM-PLS (simple), AOM-PLS (best), Ridge-default, Ridge-HPO, AOM-Ridge (simple) and AOM-Ridge (best). Counts are runner-level search units: logged trials for PLS/Ridge defaults and HPO, compact-bank CV fits for AOM-PLS, operator-alpha cells for AOM-Ridge (simple), and top-level outer-selector fits plus refits for AOM-Ridge (best).

Table 7: Software artifacts and validation status.

| Component | Tests / evidence | Status |
|---|---|---|
| `nirs4all-aom` Python package | AOM-PLS, POP-PLS, AOM-Ridge and FastAOM implementations, with unit tests and sklearn-compatible APIs. | public software |
| `nirs4all-aom` benchmark artifacts | Benchmark runners, result CSVs, aggregation scripts, figure builders and manuscript tables used here. | release with paper |
| `nirs4all` instrumentation context | NIRS instrumentation, acquisition and provenance context for local benchmark inputs. | instrumentation software |
| Supplementary validation dossier | Cohort manifest, missing-dataset audit, paired statistics and software-readiness notes distributed with `nirs4all-aom`. | public evidence |

## Supplementary Material

The supplementary material contains complete derivations, the cohort manifest, operator-bank diagnostics, non-selected AOM variants, alternative AOM-form explorations, per-dataset tables, classification details, seed stability, software artifacts and the AI-assistance statement.

## Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Anthropic Claude and OpenAI Codex to assist with code review, implementation, repository organization, benchmark aggregation scripts, LaTeX editing and drafting or revision of explanatory text. After using these tools, the authors reviewed, edited and verified the code, numerical results, references and manuscript content as needed, and take full responsibility for the content of the publication.

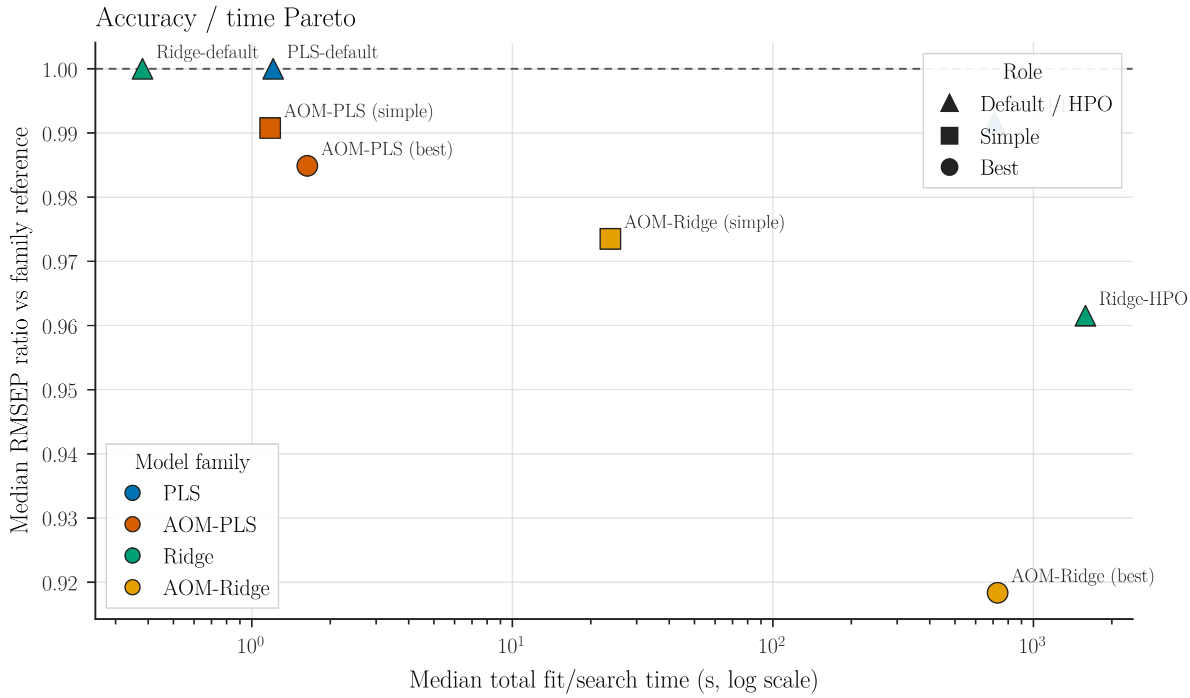


Figure 8: Accuracy/time view for all eight paper variants. The horizontal axis uses median total fit/search time: default and HPO rows include the complete CV or search/refit run, while AOM rows include the selector fit and final fitted model.

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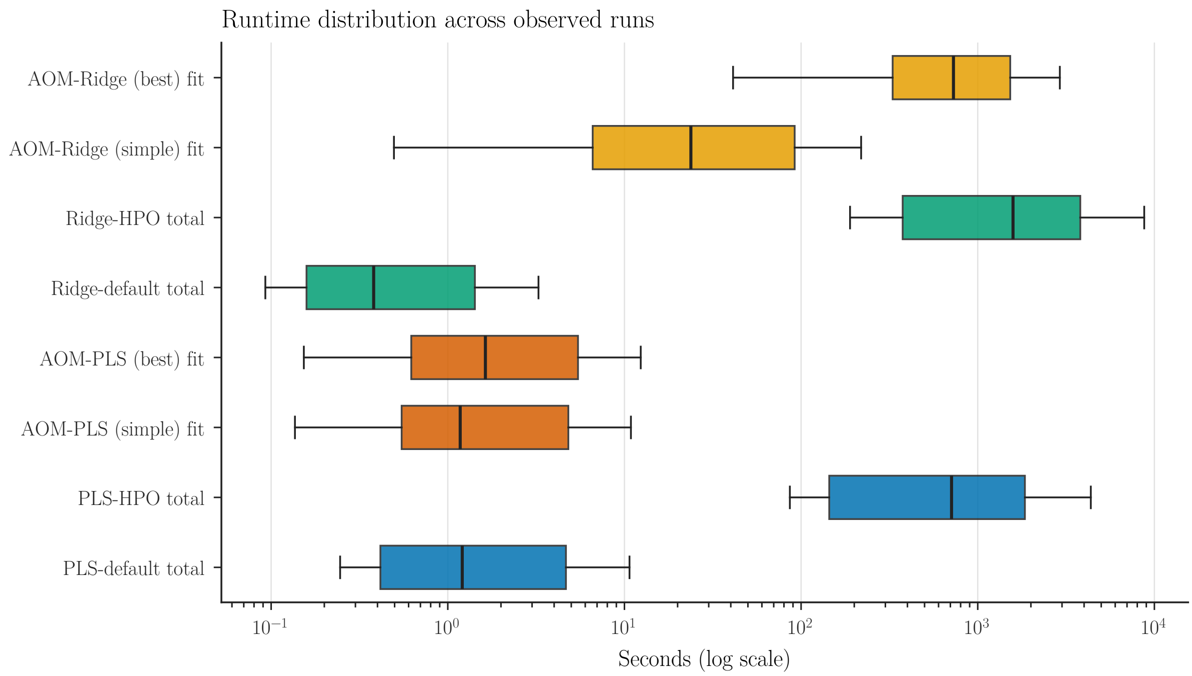


Figure 9: Runtime distributions for all eight paper variants. PLS-default, PLS-HPO, Ridge-default and Ridge-HPO boxes show total CV or HPO run time; AOM-PLS and AOM-Ridge boxes show selector fit time plus prediction overhead.


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